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ВОСХОД
КАЛУЖСКИЙ РАДИОЛАМПОВЫЙ ЗАВОД

# Four-channel analog switch 168KT2

**Brief description of the product:**

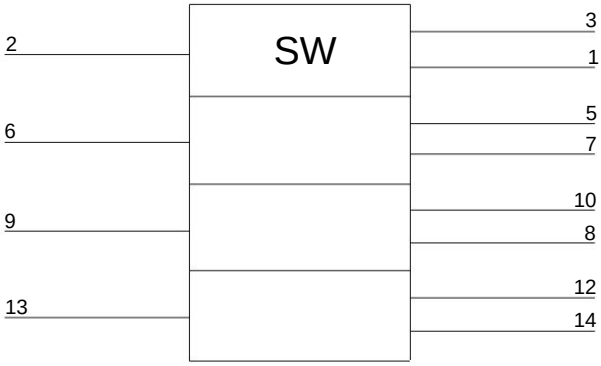
The 168KT2 IC is a four-channel analog switch designed
For switching current and voltage. It has two performance groups based on parameters (A, B, and C).
Temperature range: from -60°C to +100°C.
The microcircuit is structurally implemented in a case:
- 401.14-5.07ÿÿ (nickel, designation 168ÿÿ2ÿ1ÿÿ);
- 401.14-5M (gold, designation 168KT2TVK).
Dimensional drawing – U80.073.048GCh.
Weight – no more than 1.0 g.
Technical conditions – AEHB.43160.166TU.




The purpose of the microcircuit pins:

| Designation conclusion: | Name conclusion: |
|---|---|
| 1 | Exit |
| 2 | Control input |
| 3 | Analog input |
| 5 | Analog input |
| 6 | Control input |
| 7 | Exit |
| 8 | Exit |
| 9 | Control input |
| 10 | Analog input |
| 11 | Substrate |
| 12 | Analog input |
| 13 | Control input |
| 14 | Exit |

**Conventional graphic designation**



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## MAIN TECHNICAL CHARACTERISTICS

(at T=25±10°C)

| Parameter | Unit measurements | Norm | | | | | |
|---|---|---|---|---|---|---|---|
| | | 168KT2A | | 168KT2B | | 168KT2V | |
| | | Not less | Not more | Not less | Not more | Not less | Not more |
| Threshold voltage, B (Ui = -5B, Ii = 0.01 mA) | Upor | 3 | 6 | 3 | 6 | 3 | 6 |
| Drain leakage current, nA (Uÿ = -10V) | Iout.s | - | 20 | - | - | - | - |
| Uÿ = -15B | | - | - | - | 20 | - | - |
| Uÿ = -25B | | | - | - | - | - | 100 |
| Gate leakage current, nA (Ui = -10V) | Iout.s | - | 20 | - | - | - | - |
| Ui = -15B | | - | - | - | 20 | - | - |
| Ui = -25V | | - | - | - | - | - | 100 |
| Gate leakage current, nA (U3= -30V) Turn-on | Iout.z | - | 20 | - | 20 | - | 20 |
| time, µs. When: Ui = -5B, Usm.p = 5B | tvkl | - | 0.3 | - | 0.3 | - | 0.3 |
| Switch-off time, µs. When: Ui = -5B, Usm.p = 5B | off | - | 0.7 | - | 0.7 | - | 0.7 |
| Drain-source resistance in open state state, Ohm, at: Uÿ = -5V, Uÿ = 0V, Uÿÿ.ÿ = 5V, U3 = - 15V, Ii = 0.1 mA | Rsi.otk | - | 100 | - | 100 | - | 100 |

## ELECTRICAL PARAMETERS OF THE MICROCIRCUITS CHANGING OVER THE COURSE OF MTBF AND GAMMA PERCENTAGE LIFE.

| Parameter name, units of measurement, mode | Letter designation | Norm | | | | | | Temperature surrounding environment, C° |
|---|---|---|---|---|---|---|---|---|
| | | 168KT2A | | 168KT2B | | 168KT2V | | |
| | | Not less | Not more | Not less | Not more | Not less | Not more | |
| Drain leakage current, nA | Iut.s. | - | 50 | - | 50 | - | 200 | 25±10 |
| Drain leakage current, nA | Iout.i | - | 50 | - | 50 | - | 200 | 25±10 |
| Drain leakage current, nA | Iout.z | - | 100 | - | 100 | - | 100 | 25±10 |
| Drain-source resistance in open state, Ohm | Rsi.otk | - | 120 | - | 120 | - | 120 | 25±10 |

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**MAXIMUM PERMISSIBLE AND LIMIT MODES OF MICROCIRCUITS**

| Type denomination microcircuits | Name parameter, unit measurements | Letter designation | Norm | | Norm | |
|---|---|---|---|---|---|---|
| | | | Maximum Permissible Mode | | Limit mode | |
| | | | no less | no more | no less | no more |
| 168KT2A | The tension between drain and substrate, IN | Usp | - | 10 | - | 29 |
| 168KT2B | | | | 15 | | |
| 168KT2V | | | | 25 | | |
| 168KT2A | The tension between source and substrate, B | Uip | - | 10 | - | 29 |
| 168KT2B | | | | 15 | | |
| 168KT2V | | | | 25 | | |
| 168KT2A | The tension between shutter and substrate, B | Uzp | - | 30 | - | 35 |
| 168KT2B | | | | | | |
| 168KT2V | | | | | | |
| 168KT2A | Maximum constant switching current with one key, mA | Ikom | - | 20 | - | 23 |
| 168KT2B | | | | | | |
| 168KT2V | | | | | | |
| 168KT2A | Power dispersible microcircuit, mW | Pmax | - | 100 | - | 173 |
| 168KT2B | | | | | | |
| 168KT2V | | | | | | |

**Resistance to external influences:**

The microcircuit is resistant to external influences according to OST B 11 0998-99, including:

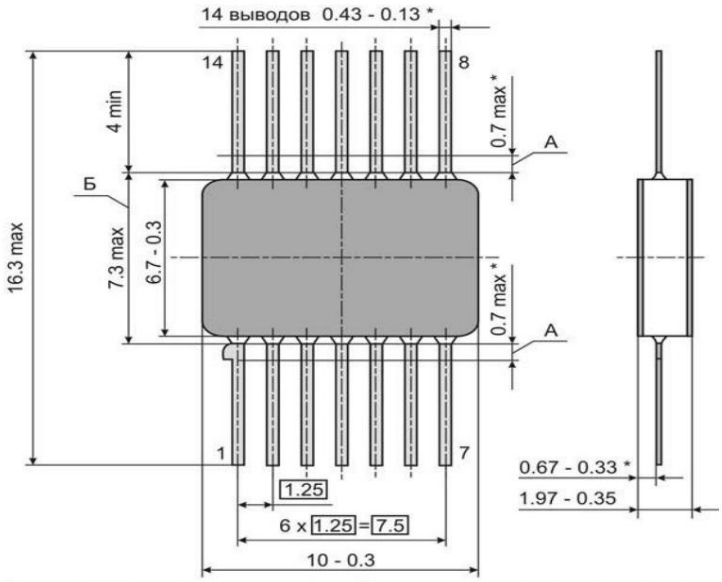
| | |
|---|---|
| Permissible value of static potential for microcircuits, V Atmospheric low working pressure, Pa (mmHg) Atmospheric high working pressure, kPa | 100<br>1.3*10-4 (10-6 ) |
| Increased relative humidity at 35°C, % Temperature change, | 294 |
| °C Linear acceleration amplitude, ms-2 (g) Sinusoidal | 98 |
| vibration: frequency range, | from - 60 to +150 |
| Hz acceleration amplitude, ms-2 (g) | 5000 (500)<br>1 ÷ 5000 |
| | 400 (40) |
| Single action mechanical strike: | |
| peak shock acceleration, ms-2 (g) duration | 15,000 (1500) |
| of action, ms | 0.1 ÷ 2.0 |
| multiple action: | |
| peak shock acceleration, ms-2 (g) duration | 1500 (150) |
| of action, ms | 1 ÷ 5 |

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**Requirements for resistance to special factors:**

| Types of special factors | Characteristics of special factors factors according to GOST RV 20.39.414.2 | Values of characteristics special factors |
|---|---|---|
| 7.And | 7.I1 | 1Us |
| | 7.I6 | 1Us |
| | 7.I7 | 1Us |
| | 7.I8 | 4.5x10-7 x1Us |
| 7.C | 7.C1 | 1Us |
| | 7.C4 | 0.1x1Us |
| 7.K | 7.K1 | 0.05x1K |
| | 7.K4 | 0.05x1K |

**General appearance, pinout diagram and overall dimensions of the case:**




А - длина вывода, в пределах которой производится контроль смещения плоскостей симметрии выводов от номинального расположения.
Б - ширина зоны, которая включает действительную ширину микросхемы и часть выводов, непригодную для монтажа.

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